INCH - POUND
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PERFORMANCE SPECIFICATION

CLOTH, FLAME RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for flame resistant (FR) cloth, for use in flame resistant clothing.

1.2 Classification. This specification covers the following five (5) types and classes.

1.2.1 Types.

Type I - Flame resistant, 65 percent rayon/25 percent para-aramid/10 percent nylon, ripstop
Type II - Flame resistant, 43 percent meta-aramid/30 percent nylon/27 percent cotton, twill
Type III - Flame resistant cloth, ripstop
Type IV - Flame resistant cloth, twill
Type V - Flame resistant antistatic cloth, twill

1.2.2 Classes. (see 6.10).

Class 1 - Operational Camouflage Pattern (OCP)
Class 2 - Coyote 498
Class 3 - Digital Print Desert Camouflage (Navy Working Uniform (NWU) II
Class 4 - Digital Print Woodland Camouflage (Navy Working Uniform (NWU) III

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil/.

AMSC N/A FSC 8305

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.–The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY, DEVCOM – SOLDIER CENTER

2-1-2592 - Operational Camouflage Pattern (OCP)

(Copies of drawings are available from the U.S. Army, DEVCOM Soldier Center, ATTN: FCDD-SCP-PEI, 10 General Greene Avenue, Natick, MA 01760-5019.)

NAVY CLOTHING AND TEXTILE RESEARCH FACILITY

01-NCTRF-12 - Desert Camouflage NWU II
02-NCTRF-12 – Woodland Camouflage NWU III

(Copies of these documents are available from the procuring activity.)

OTHER GOVERNMENT DOCUMENTS AND PUBLICATIONS

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CODE OF FEDERAL REGULATIONS

Title 16 CFR Part 303 – Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP2 - Evaluation Procedure for Gray Scale for Staining
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter Method
AATCC TM15 - Test Method for Colorfastness to Perspiration
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon Arc
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM20A - Test Method for Fiber Analysis: Quantitative
AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated
AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM116 - Test Method for Colorfastness to Crocking, Rotary Crockmeter Method
AATCC TM135 - Test Method for Dimensional Changes of Fabrics after Home Laundering
AATCC TM143 - Test Method for Appearance of Apparel and Other Textile End Products After Repeated Home Laundering

(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1424 - Standard Test Method for Tearing Strength of Woven Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D3512/D3512M - Standard Test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Random Tumble Pilling Tester

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ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M - Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Grab Test)
ASTM D6413/D6413M - Standard Test Methods for Flame Resistance of Textiles Vertical Test

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Basic material.

3.4.1 Fiber content.

3.4.1.1 Type I cloth. Type I cloth shall be made of warp and filling yarns using a fiber blend ratio of 65 percent flame resistant rayon, 25 percent para-aramid and 10 percent nylon.

3.4.1.2 Type II cloth. Type II cloth shall be made of warp and filling yarns using a fiber blend ratio of 43 percent meta-aramid, 30 percent nylon and 27 percent cotton.

3.4.1.3 Type III and Type IV cloths. Types III and IV cloths shall be made of warp and filling yarns which result in a flame resistant cloth that meets the requirements of Table III, except the contracting authority approval shall not apply to Class 3 and Class 4 cloths.

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3.4.1.4 Type V cloth. Type V cloth shall be a blend of inherently flame resistant fibers, synthetic fibers, cellulosic fibers, and static-dissipative fibers which result in a flame-resistant cloth that meets the requirements of Table III and has received approval by the Contract Authority. Both the warp and filling yarns shall contain two (2) percent minimum static-dissipative fibers based upon the dry weight of the desized cloth. Testing shall be as specified in 4.5.

3.4.2 Weave.

3.4.2.1 Type I and Type III. The weave for Type I and Type III shall be plain weave with reinforcement ribs in both the warp and filling directions. The ribs for Type I and Type III should be formed by weaving two (2) warp ends as one (1) and two (2) picks as one (1) in the same shed, in an interval such that achieves the performance requirements of this specification.

3.4.2.2 Type II and Type IV. The weave for Types II and IV shall be 3 over 1 or 2 over 1 left-hand twill weave.

3.4.2.3 Type V. The weave for Type V shall be a 3 over 1 left-hand twill weave.

3.5 Color.

3.5.1 Visual shade matching (all Classes). The color and appearance of the dyed and camouflage printed cloth(s) shall match the standard sample when tested as specified in 4.5.

3.5.1.1 Class 1, Operational Camouflage Pattern (OCP). The finished cloth shall be either dyed to a ground shade matching Dark Cream 559 and then overprinted with the remaining six (6) colors Tan 525, Light Sage 560, Olive 527, Dark Green 528, Brown 529, and Bark Brown 561 areas of the camouflage pattern or when the ground shade is not dyed to Dark Cream 559 all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern.

3.5.1.2 Class 2, Coyote 498. The cloth shall be dyed to match the standard sample for Coyote 498 (see 6.4).

3.5.1.3 Digital print desert camouflage NWU II – (Class 3). The NWU II printed cloth shall be dyed to a ground shade either matching or approximating Desert Tan 511 and then shall be printed with the camouflage pattern. When the ground shade is dyed to match Desert Tan 511, the three (3) remaining colors shall be printed as appropriate for the Urban Tan 512, Light Brown 513 and Dark Brown 514 areas of the pattern. As an alternate, when the ground shade is not dyed to approximate Desert Tan 511, all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors. Each area of the specific color of the pattern shall be in accordance with the applicable standard sample or drawing number 01-NCTRF-12.

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3.5.1.4 Digital print woodland camouflage NWU III – (Class 4). The NWU III printed cloth shall be dyed to a ground shade either matching or approximating Tan 516 and then shall be printed with the camouflage pattern. When the ground shade is dyed to match Tan 516, the three (3) remaining colors shall be printed as appropriate for the Olive 517, Green 518 and Black 519 areas of the pattern. As an alternate, when the ground shade is not dyed to approximate Tan 516, all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors. Each area of the specific color of the pattern shall be in accordance with the applicable standard sample or drawing number 02-NCTRF-12.

3.5.2 Colorfastness. The finished cloth shall conform to the colorfastness requirements listed below in Table I, Table I.A, Table I.B and Table I.C when tested in accordance with the methods specified in 4.5 and 4.6.2.

TABLE I Colorfastness requirements, Class 1, all types.

Operational Camouflage Pattern (OCP)				
Colors Evaluation	Laundering (1 cycle) (color change and staining) (min.)	Light (40 AFU or 170 kJ /(m^2nm) @ 420 nm) (min.) 1/	Perspiration (color change and staining) (min.)	Crocking Wet and Dry (min.)
All colors	-	-	3-4	-
Dk. Green 528, Brown 529, Bark Brown 561	2-3	3-4	--	Wet 2 Dry 3.5
Dark Cream 599, Tan 525, Light Sage 560, Olive 527	3	3	--	Wet 3 Dry 4

1/ AFU. AATCC Fading Units

TABLE I.A. Colorfastness requirements, Class 2, all types.

Coyote 498				
Colors Evaluation	Laundering (1 cycles) (color change and staining) (min.)	Light (40 AFU. or 170kJ/(m^2nm) @ 420 nm) (min.)	Perspiration (color change and staining) (min.)	Crocking Wet and Dry (min.)
Coyote 498	3-4	3	3-4	Wet 2.5, Dry 3

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TABLE I.B. Colorfastness requirements, Class 3 all types.

Colors Evaluation	Laundering (min.)	Light (min.)	Perspiration (min.)		Crocking (min.)	
	Color Change and staining (1 cycle)	40 AFU or 170 kJ/(m^{2}nm)@ 420 nm 1/	Color Change and Staining		Dry	Wet
All Colors	-	-	3.5	3.5	-	-
Desert Tan 511 Urban Tan 512 Light Brown 513	2-3	3	-	-	4	3
Dark Brown 514	2-3	3-4	-	-	3.5	2.5

1/ AATCC Fading Units (AFU)

TABLE I.C. Colorfastness requirements, Class 4 all types.

	Laundering (min.)	Light (min.)	Perspiration (min.)		Crocking (min.)	
	Color Change and staining (1 cycle)	40 AFU or 170 kJ/(m^{2}nm)@ 420 nm 1/	Color Change and Staining		Dry	Wet
All Colors	-	-	3.5	3.5		
Tan 516 Olive 517 Green 518	2-3	3	-	-	4	3
Black 519	-	3-4	-	-	-	-

1/ AATCC Fading Units (AFU)

3.5.3 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed cloth shall be "Free" of labile sulfur when tested as specified in 4.5 and 4.6.3.

3.6 Pattern execution (Class 1). The pattern on the printed finished cloth(s) shall match the standard sample in respect to design, colors and registration of the respective areas. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations and/or feathering as required for each color. Each pattern area shall show full coverage; skitteriness, feathering, haloing or trapping and off register exceeding that shown on the standard sample in any of the printed areas will not be acceptable. When the standard sample is not available for pattern execution, a pattern drawing will be provided (see 6.4) and the pattern on the finished cloth shall match that of the drawing 2-1-2592 when tested as specified in 4.5. The pattern repeat for Class 1 shall be as follows:

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Class 1 - Operational Camouflage pattern (OCP)	- 25.255 (+1.25, -2.50) inches in the warp direction and 68-inches maximum camouflage printed area in the filling direction.

3.6.1 Pattern execution (Class 3). The pattern of the printed finished cloth(s) shall match the standard sample in respect to design, colors and registration of the respective areas. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations as required for each color. Each pattern area shall show full coverage; skitteriness, feathering, haloing, or trapping and off register exceeding that shown on the standard sample in any of the printed areas shall not be acceptable. When the standard sample is not available for pattern execution, a pattern drawing shall be provided (see 6.4) and the pattern on the finished cloth shall match that of the drawing 01-NCTRF-12 when tested as specified in 4.5. The Anchor, Constitution and Eagle (ACE) logo shall be utilized on all Class 3 material. The ACE logo shall be Light Brown 513 on Urban Tan 512. The pattern repeat for Class 3 shall be as follows:

Class 3- Digital Print Desert Camouflage NWU II	36.00 (±1.5,) inches in the warp direction 33.00 (±1.5) inches in the filling direction

3.6.2 Pattern execution (Class 4). The pattern of the printed finished cloth(s) shall match the standard sample in respect to design, colors and registration of the respective areas, The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations as required for each color. Each pattern area shall show full coverage; skitteriness, feathering, haloing, or trapping and off register exceeding that shown on the standard sample in any of the printed areas shall not be acceptable. When the standard sample is not available for pattern execution, a pattern drawing shall be provided (see 6.4) and the pattern on the finished cloth shall match that of the drawing 02-NCTRF-12 when tested as specified in 4.5. The Anchor, Constitution and Eagle (ACE) logo shall be utilized on all Class 4 material. The ACE logo shall be Green 518 on Olive 517. The pattern repeat for Class 4 shall be as follows:

Class 4-Digital Print Woodland Camouflage NWU III	36.00 (±1.5) inches in the warp direction 36.00 (±1.5) inches in the filling direction

3.8 Spectral reflectance.

3.8.1 Class 1, Operational Camouflage Pattern (OCP). The spectral reflectance values shall conform to the requirements specified in Table II, when tested as specified in 4.6.4.

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TABLE II. Spectral reflectance requirement. Class 1, all types (percent).

Operational Camouflage Pattern (OCP)						
Wavelength, Nanometers (nm)	Dark Cream 559 Tan 525		Light Sage 560 Olive 527, Brown 529		Dark Green 528 Bark Brown 561	
	Min.	Max.	Min.	Max.	Min.	Max.
600	22	44	12	30	3	11
620	24	45	12	30	3	12
640	24	45	12	32	4	13
660	25	45	12	32	4	14
680	28	45	14	34	4	17
700	28	48	14	36	6	23
720	30	52	16	39	6	23
740	32	55	18	41	10	25
760	36	56	20	43	14	30
780	38	57	22	45	18	35
800	40	57	22	45	21	40
820	44	58	24	46	24	42
840	46	59	26	47	26	43
860	48	60	28	48	28	45

3.8.2 Class 2, Coyote 498. The spectral reflectance of the color in the Coyote 498 cloth shall conform to the requirements specified in Table II.A, when tested as specified in 4.6.4.

TABLE II.A. Spectral reflectance requirement, Class 2, all types (percent).

Coyote 498		
Wavelength, Nanometers (nm)	Min.	Max.
600	8	20
620	8	20
640	8	22
660	8	24
680	12	24
700	12	34
720	16	42
740	22	46
760	30	50
780	34	54
800	36	56
820	38	58
840	38	58
860	40	60

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3.8.3 Class 3, Digital Print Desert Camouflage NWU II. The spectral reflectance shall conform to the requirements specified in Table II.B, when tested as specified in 4.6.4.

TABLE II.B. Spectral reflectance requirement, Class 3, all types (percent).

Wavelength, Nanometers (nm)	Desert Tan 511		Urban Tan 512 & Light Brown 513		Dark Brown 514	
	Min.	Max.	Min.	Max.	Min.	Max.
700	32	50	27	33	7	26
720	33	51	28	38	7	26
740	33	51	26	40	7	26
760	35	52	24	40	7	28
780	36	53	26	42	7	28
800	37	54	27	42	7	28
820	38	54	28	43	7	30
840	39	55	29	44	7	30
860	40	56	30	45	7	30

3.8.4 Class 4, Digital Print Woodland Camouflage NWU III. The spectral reflectance shall conform to the requirements specified in Table II.C, when tested as specified in 4.6.4.

TABLE II-C Spectral reflectance requirement, Class 4, all types (percent).

Wavelength, Nanometers (nm)	Tan 516		Olive 517 & Green 518		Black 519	
	Min.	Max.	Min.	Max.	Min.	Max.
600	23	40	10	23	0	10
620	23	40	9	23	0	10
640	24	42	9	24	0	10
660	25	44	9	25	0	12
680	28	46	10	30	0	14
700	33	51	11	39	0	20
720	40	55	14	42	5	23
740	45	62	21	47	6	24
760	51	71	28	50	7	24
780	53	75	33	52	8	25
800	53	75	38	52	9	28
820	54	77	40	54	10	28
840	55	77	42	55	11	29
860	56	78	43	58	11	30
880	58	81	45	60	12	31
900	60	83	46	60	12	32

3.9 Physical requirements. The finished cloth(s) shall conform to the physical requirements listed below, in Table III, when tested in accordance with the test methods specified in 4.5.

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TABLE III. Physical requirements.

Characteristic	Requirement				
	Type I	Type II	Type III	Type IV	Type V
Weight, oz./sq.yd.					
Min.	5.5	5.5	6.0	5.7	5.75
Max.	8.5	8.5	7.0	7.2	6.75
Breaking strength, pounds (min.)					
Warp	100	100	110	110	150
Filling	80	80	100	100	100
Tearing strength, pounds (min.)					
Dry Warp	4.0	4.0	12.0	12.0	10.0
Filling	4.0	4.0	10.0	10.0	12.0
Wet Warp	N/A	N/A	8.0	8.0	N/A
Filling	N/A	N/A	6.0	6.0	N/A
Dimensional stability, percent, (max.) After laundering (Warp and Filling) (1 cycle and 5 cycles)					
Individual Sample	4.0	5.5	4.0	5.5	5.0
Lot Average	3.5	5.0	3.5	5.0	4.0
Air permeability, $ft^3/min/ft^2$ (min.)	10.0	10.0	25.0	25.0	25.0
Flame resistance, initial:					
After flame, seconds (max.)	2.0	2.0	2.0	2.0	2.0
After glow, seconds (max.)	25.0	25.0	15.0	15.0	10.0
Char length, inches (max.)	4.5	4.5	4.5	4.5	5.0
Flame resistance, after laundering: (25, 50 and 100 cycles)					
After flame, seconds (max.)	2.0	2.0	2.0	2.0	2.0
After glow, seconds (max.)	25.0	25.0	15.0	15.0	10.0
Char length, inches (max.)	4.5	4.5	4.5	4.5	5.5
Flash Fire (Predicted Burn Injury) 1/ Percent, (max.)					
With head	37	37	35	35	35
Without head	30	30	28	28	28
Fabric appearance/smoothness, rating (min.)					
Initial	5	5	5	5	5
After laundering (15 cycles)	4	4	3	3	3
Pilling, 30 minutes (min.)	N/A	N/A	N/A	N/A	3.0
Electrostatic decay, seconds (max.) voltage level +/- 4000 volts (min.)					
Initial	N/A	N/A	N/A	N/A	0.5
After laundering (5 cycles)	N/A	N/A	N/A	N/A	0.5

1/ This requirement and testing is performed according to the documentation for the end item specified in the document or solicitation during first article testing only.

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3.10 pH. The pH of the water extract of the finished cloths shall be not less than 5.0 and not greater than 8.5 when tested as specified in 4.5.

3.11 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.6.6.

3.12 Face identification. The warp side shall be identified as the face side by stamping that side with the word "FACE" at each end of the roll.

3.13 Width. For Government procurements only, the width of the finished cloths shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

3.14 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put-up full width on a roll as specified in 5.1.

3.15 Fiber identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with Title 16 CFR Part 303 Rules and Regulations under the Textile Fiber Products Identification Act.

3.16 Workmanship. The finished cloths shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1).
b. Conformance inspection (see 4.3.2).

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified, or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. The first article, submitted in accordance with 3.1 shall be inspected for appearance, color, the defects of 4.4 and tested for the characteristics as specified in 4.5 through 4.6.6.

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4.3.2 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.6.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits as specified in the contract and/or procurement documents, except where otherwise indicated (see 6.2).

4.4 Visual examination. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with Type I of MIL-STD-3064.

4.4.1 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The cloth shall be tested for the characteristics listed in Table IV. The methods of testing as specified wherever applicable and as listed in Table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. End item tests.

Characteristic	Requirement Reference	Test
Fiber identification	3.4.1	AATCC TM20 (see 6.8)
Fiber content	3.4.1	AATCC TM20A (see 6.8)
Weave	3.4.2	Visual
Visual shade matching	3.5.1	4.6.1
Colorfastness to: Laundering Light Perspiration Crocking	Table I, I.A, I.B, I.C	4.6.2 AATCC TM61 3A 1/, 2/ AATCC TM16.3, Opt 3 3/ AATCC TM15 1/, 2/ AATCC TM116, 4/ or AATCC TM8 (Class 2 only), 4/
Presence of labile sulfur	3.5.3	4.6.3 5/ (see 6.7)

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TABLE IV. End item tests. - Continued

Characteristic	Requirement Reference	Test
Pattern execution	3.6	Visual and linear measurement using a Precision Steel Ruler
Spectral reflectance	Table II, II.A, II.B, II.C	4.6.4
Weight	Table III	ASTM D3776/D3776M (Method C)
Yarns per inch	Table III	ASTM D3775
Breaking strength	Table III	ASTM D5034
Tearing strength	Table III	ASTM D1424
Dimensional stability (after 1 & 5 laundering cycles)	Table III	AATCC TM135, (3)(V)Aiii
Air permeability	Table III	ASTM D737
Flame resistance Initial, After laundering (25, 50 and 100 cycles)	Table III	ASTM D6413/D6413M AATCC TM135, (3)(V)Aiii and ASTM D6413/D6413M 5/
Fabric appearance/smoothness: Initial After laundering (15 cycles)	Table III	AATCC TM143 6/ AATCC TM135, (3),(V),Aiii & AATCC TM143 6/
Flash Fire (predicted burn injury) with and without head	Table III	7/
Pilling (30 minutes)	Table III	ASTM D3512/D3512M
Electrostatic decay Initial After laundering (5 cycles)	Table III	4.6.5 AATCC TM135 (3),(V),(Aiii) and 4.6.6
pH	3.10	AATCC TM81
Toxicity	3.11	4.6

1/ Only the stain on the nylon and cotton fibers of the color transfer cloth shall be evaluated.
2/ Rated using the AATCC EP1, and AATCC EP2.
3/ Rated using the AATCC EP1.
4/ Rated using the AATCC EP8.
5/ Tests after 50 and 100 cycles are required for First Article only. Tests after 25 cycles required for lot testing.
6/ Rated using the AATCC Smoothness and Appearance Replicas.
7/ Material in applicable type and class shall be tested in accordance to applicable garment end item specification. For two-piece uniforms, the coat, or shirt, shall be tucked into the waist of trouser with a 100% Cotton T-shirt and brief underneath the uniform. Manufacturer shall submit three (3) sets of garments to Product Manager Soldier Clothing and Individual Equipment (PdM-SCIE) for all Class 1, except for Class 2 where, three (3) specimens will be sent to Product Manager Infantry Combat Equipment (PdM-ICE) for testing.

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4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the finished cloth shall match the standard sample when viewed using the AATCC EP9, Option C, (see 6.5) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the finished cloth shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.5.1). The finished cloth shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

4.6.2 Colorfastness evaluation testing of printed cloth. When testing for colorfastness properties on printed cloth, each color shall be evaluated, whenever possible, separately and reported as such. In cases where the print pattern does not allow for the evaluation of each color separately, the test results should indicate which colors were evaluated together.

4.6.2.1 Operation Camouflage pattern (OCP). All colorfastness testing for Operation Camouflage Pattern shall be performed on the solid color area and not the tonal area.

4.6.3 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g SnCl2 (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail.

4.6.4 Spectral reflectance test. Spectral reflectance data for Class 1 and Class 2 shall be obtained from 600 to 860 nanometers (nm) at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Spectral reflectance data for Class 3 shall be obtained from 700 to 860 nanometers (nm) at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of

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compounds, the preferred white standard. Spectral reflectance data for Class 4 shall be obtained from 600 to 900 nanometers (nm) at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6-inches away from the selvage. The specimen shall be measured as a single layer backed with four (4) layers of the same fabric and shade. The specimen shall be viewed at an angle no greater than 10° from normal, with the specular component included. Measurements shall be taken on a minimum of two (2) different areas. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same Warp or filling yarns/ Wales or courses when presented to the sample port. Camouflage materials should be measured with the appropriate aperture size to ensure that only one (1) color is measured at a time. Always use the largest aperture possible. Photometric accuracy of the spectrophotometer shall be within one (1) percent and wavelength accuracy within two (2) nm. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.6.5 Electrostatic decay test (Type V only). The specimen shall be tested according to the method specified in 4.6.5.1 through 4.6.5.3 initially and after five (5) laundering cycles.

4.6.5.1 Specimen preparation. The specimen shall be a 3-inch by 6-inch rectangular piece of material. The warp or filling yarns shall run parallel to the 6-inch length of the specimen. The direction of test (warp or filling) shall be along the length of the specimen and each specimen shall be labeled accordingly in one (1) corner. Test specimens shall be cut so that no two (2) contain the same set of warp and filling yarns. Six (6) specimens, three (3) warp and three (3) filling, shall be tested, face and back sides inclusive.

4.6.5.2 Testing equipment. The test equipment shall consist of a meter for measuring relative humidity (RH) and temperature, a faraday cage, a source capable of outputting 5000 volts, and voltage meters to display the applied voltage and the voltage detected by the sensor. The faraday cage shall contain two (2) parallel electrodes on which specimens may be mounted and charged, and a sensor that will detect voltage on the specimen surface. The faraday cage and specimens to be tested shall be pre-conditioned at 10 (± 2) percent relative humidity overnight and conditioned at 20 (± 2) percent relative humidity for a minimum of 24 hours at approximately 75º (± 5)°F. The specimens shall be tested at 20 (± 2) percent relative humidity and 75º (± 5)°F. A bench top size chamber may be used as an option to achieve the conditions and to do the testing in, or any area capable of maintaining these conditions may be used. An air ionizing blower may be used while conditioning, although the blower must be turned off once testing begins. A static master may also be used to eliminate residual charges before testing.

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4.6.5.3 Testing procedure. Allow the test equipment 1-hour of warm-up time. Adjust the voltage source for a charging voltage of 5000 volts by mounting a 3-inch by 5-inch aluminum plate across the electrodes in the faraday cage such that it is centered in front of the voltage sensor opening. Apply a charge of 5000 volts and adjust/calibrate the voltage meter designated to indicate detected voltage on the sample equal to 5000 volts. Remove the aluminum plate. Mount a specimen tautly across the electrodes, centering it over the sensor opening. The surface of test specimen (back or front) shall face the sensor. The CUT-OFF must be set to 10 percent. With the high voltage off, depress the CHG button to measure initial charge on the test specimen. If it is not reading zero, turn CHG button off and use the static master or blower to attempt to eliminate the residual charge. With high voltage off, depress the CHG button again and record any remaining measurable residual charge. Turn off CHG button. Apply 5000 volts to the electrodes for a period of 20 seconds. To do this, depress the + button on the right (the reading on CHARGING VOLTAGE meter must read 5000 volts); then depress the CHG button. Wait 20-second; record the highest reading on the SAMPLE CHARGE meter. Then depress the TEST button and from the readout under DECAY TIME-SEC, record how long it took for the charge to decay from maximum voltage to 10 per cent of the maximum level. Depress the ZERO/STBY button after the test. The high voltage (5000 volts) shall be turned off and the specimen immediately grounded. If the DECAY TIME-SEC is less than 0.5 seconds, the specimen is acceptable; otherwise, it is not. Also, the specimen must accept a charge of at least 4000 volts during the 20 second charging period to be acceptable.

NOTE. The reason for failure shall be noted. Recalibrate periodically, at the minimum between each set of six (6) specimens. Report the test method used and any alterations to the method shall be reported. Test conditions, including relative humidity, temperature and conditioning time shall be stated (see 6.6).

4.6.6 Toxicity test. When the toxicity requirement (see 3.11) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required, (see 6.7 and 6.8).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. The cloth is intended for use in flame resistant, camouflage clothing.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type and class required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1, 4.2, 6.3)
e. Conformance inspection acceptance quality limits (AQLs)(see 4.3)
f. Camouflage pattern drawing, if required (see 2.2.2 and 6.4)
g. When toxicity testing is required (see 3.11, 4.6.6)
h. Width of cloth required (see 3.12)
i. Length required if other than specified (see 3.13)
j. Inspection conditions (see 4.3.1)
k. Flash Fire garments testing required (see 4.5 and 6.3)
l. Packaging (see 5.1)

6.3 First article. When a first article inspection is required (see 3.1.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens".

6.5.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75.

6.6 Electrostatic decay testing equipment known supplier: Electro-Tech Systems Inc., 115 E. Glenside Avenue, Glenside PA 19038.

6.7 Toxicity testing. If dermal toxicity testing is required, it will be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1)

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Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100)

Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250)

6.7 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.8 Superseding documents. This document supersedes GL-PD-07-12 Cloth, Flame Resistant (all revisions) and will be used in all future procurements.

6.9 Classes. Previous classes cited in GL-PD-07-12 Cloth, Flame Resistant have been changed to the following in this specification:

GL-PD-07-12	**MIL-PRF-32635**
Class 1- Universal Camouflage Pattern (UCP)	Class 1-Operational Camouflage Pattern (OCP)
Class 2-DELETED	Class 2-Coyote 498
Class 3- Operational Camouflage Pattern (OCP)	
Class 4-Coyote 498	

NOTE. Universal Camouflage Pattern (UCP) wear out date was September 30, 2019, therefore it was not included in the Military specification.

6.11 Changes since last revision. Class 3 and Class 4 were added for the United States Navy.

6.12 Subject term (key word) listing.

Coyote 498
Digital Print Desert Camouflage, Navy Working Uniform (NWU II)
Digital Print Woodland Camouflage NWU III
Dissipative Fiber
Operational Camouflage Pattern (OCP)

Custodians:
Army - GL
Navy- NU
Air Force - 11

Preparing activity:

DLA-CT

(Project 8305-2024-001)

NOTE: The activities listed above were interested in this document as of date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST online database at https://assist.dla.mil.

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